Department of Mechanical Engineering, Punjabi University, Patiala

Engineering Graphics, Second Mid-Semester Exam, April, 2024

Arnav Garg
1180

B. Tech. Ist Year, 1st Sem (CSE)

Time: 1 Hour

Max. Marks: 15

NOTE: Section A is compulsory. Attempt any two questions from Section B.

Q.1	**Section A: Answer the following five questions with pen only.**	
(i)	Give two examples of Solids of revolution. axles, funnels, Cylinder.	01
(ii)	The Truncated solids are the solids cut by cutting plane ________ to its base. (parallel / perpendicular / inclined)	01
(iii)	An oblique solid has its axis ________ to its base. (parallel / perpendicular / inclined)	01
(iv)	A square pyramid is resting on its base is cut by a cutting plane parallel to VP and perpendicular to HP. The true shape of the cut section will be visible in ________ view. (Front /Top / Side view)	01
(v)	The isometric dimension is ___81.5___ % of the true dimension.	01
	Section B: Draw any two out of the following three questions.	
Q. 2	A hexagonal prism of base 25 mm side and axis 45 mm long is positioned with one of its base edges on H.P. such that the axis is inclined at 30° to H.P. and plan of the axis inclined at 45° to the V.P. Draw its projections of the prism.	05
Q. 3	A cylinder of 40 mm diameter, 60 mm height and having its axis vertical, is cut by an AIP section plane perpendicular to the V.P., inclined at 45° to the H.P. and intersecting the axis 32 mm above the base. Draw its front view, sectional top view and true shape of the section.	05
Q. 4	A Pentagonal prism of edge of 25 mm and axis length of 55 mm is resting on its rectangular face on ground with its axis parallel to both H.P and V.P. Draw its isometric projection.	05

1210 1078
Prem Kumar

Department Of Mechanical Engineering, Punjabi University, Patiala

END SEMESTER EXAMINATION JUNE-2022

ENGINEERING GRAPHICS

(B.Tech 1ST YEAR, GROUP A)

TIME ALLOWED: 2 HOURS **MAX. MARKS: 50**

NOTE: SECTION A IS COMPULSORY. ATTEMPT ANY THREE QUESTIONS FROM SECTION B.

Q.1	**Section A** ***All parts of this question are compulsory*** **Marks = (2*10= 20)**
(i)	In orthographic projections, the projectors are drawn from an object are ________ to each other and ________ to the plane of projection. (Parallel, Inclined, perpendicular)
(ii)	Draw symbols of first and third angle projection systems.
(iii)	A straight line is inclined to HP and parallel to VP. The ______ view of the line shows the true length and ______ view shows the apparent length of the line. (Front, Side, Top)
(iv)	The cube is a solid that falls in the category of ______. (Prism, Pyramid, Regular Poyhedra)
(v)	A point A is in VP and 10 mm below HP. The point lies in ________ quadrant(s). (I, II, III, IV)
(vi)	A plane lamina is perpendicular to VP and inclined to HP. The front view of the lamina shows ________ and the top view shows ______ (True Shape, Edge View, Apparent shape)
(vii)	In an isometric projection, the three axes are inclined at ______ to each other. (45°, 90°, 120°)
(viii)	A solid is resting on HP with its axis perpendicular to it. It is cut by a cutting plane parallel to VP. The front view shows the ______ of the cut solid. (True shape, Apparent shape, Cutting Plane Line)
(ix)	The radial line method is used for the development of ______. (Cylinder, Cone, Square prism)
(x)	What is a skew line.
	Section B (10 x 3 = 30) Attempt any three questions.
Q. 2	A line AB 80 mm long has its end A is 40 mm above HP and 15 mm in front of VP. The end B in 10 above HP and 70 mm in front of VP. Draw the projections of the line and find the inclinations of the line with HP and VP. Also locate its HT and VT.
Q.3	A right regular pentagonal pyramid of base side 30 mm and height 60 mm long is resting with one of its base sides in HP. The axis of the solid is inclined at 30° to the HP and side of the base on which it rests in HP is inclined at 45° to VP. Draw the projections of the pyramid. OR A circular lamina of diameter 50 mm rests on a point of its circumference in HP with its surface inclined at 35° to HP. The plan of diameter passing through the point is inclined at 50° to VP. Draw the projections of the lamina.
Q. 4	A right regular hexagonal pyramid of base side 40 mm, height 80 mm is resting on its base in HP with an edge of the base perpendicular to VP. It is cut by a cutting plane perpendicular to VP and inclined at 45° to HP bisecting the axis. Draw the front view, sectional top view and obtain its lateral development.
Q.5	Draw the front view, left hand side view and top view of the object shown in Figure 1.
Q. 6	A right square pyramid lies centrally on the top of rectangular surface of a hexagonal prism, as shown in Figure 2. Draw the isometric view of the combined solid.

a' above
a in
a behind
below
a'

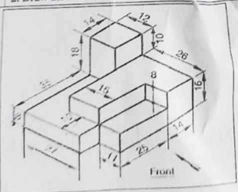

Figure 1

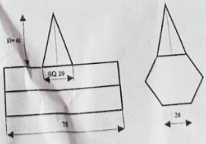

Figure 2

DEPARTMENT OF MECHANICAL ENGINEERING

PUNJABI UNIVERSITY, PATIALA

END SEMESTER EXAMINATION, FEBRUARY-2022

B.Tech 1ST YEAR, GROUP B

MCE-151, ENGINEERING GRAPHICS

TIME ALLOWED: 2 HOURS **MAX. MARKS: 50**

NOTE: SECTION A IS COMPULSORY. ATTEMPT ANY THREE QUESTIONS FROM SECTION B.

	Section A
Q. 1	***All parts of this question are compulsory*** **Marks (2*10= 20)**
I.	A point is 30 mm below HP and in VP. The point lies in __3rd__ and __4th__ quadrants.
II.	A prism is solid having __Similar__ end faces joined together by __Rectangular__ surfaces. (Dissimilar, Similar: Rectangles, Triangles).
III.	What is the difference between the frustum of a solid and a truncated solid?
IV.	A line is parallel to VP and perpendicular to HP. Its point view will be obtained on the __Horizontal__ plane.
V.	A straight line AB is inclined to HP and VP. The side view of the line shows __________. (True Length, Point View, Projected Length, None of these)
VI.	Differentiate between an apparent and a true section.
VII.	In third angle projection, the top view of an object lies __Above__ the front view. (Above, Below, on XY Line)
VIII.	A plane lamina is perpendicular to HP and inclined to VP. The __Top__ view of the lamina shows an edge view.
IX.	The development of a right circular cylinder is obtained by Parallel Line __Development__ method.

The lines drawn parallel to the isometric axes are known as Isometric Line. Any other line which is not parallel to any of the isometric axes is known as Non-Isometric Line. The lines XY, YZ & ZX are called non-isometric lines.

X.	What is the difference between an isometric and non-isometric line.
	Section B (10 x 3 = 30) Attempt any three questions.
Q. 2	A line AB 90 mm long is inclined at 45° to HP and its top view makes an angle of 60° with XY line. The end A is in HP and 12 mm in front of VP. Draw its projections and find its true inclination with VP. Locate HT and VT.
Q. 3	A pentagonal pyramid of base side 40 mm and height 80 mm rests with its base in HP with an edge of the base making 30° to VP. It is cut by a section plane perpendicular to VP and inclined at 40° to HP bisecting the axis. Draw the front view, sectional top view and true shape of the section.
Q. 4	A right square pyramid of base side 40 mm and height 80 mm is resting on HP on its base with one of the base edges inclined at 30° to VP. It is cut by a cutting plane perpendicular to VP and inclined at 45° to HP, bisecting the axis of the solid. Draw elevation, sectional plan and develop the lateral surface of the solid.
Q. 5	A hexagonal prism of base side 30 mm and height 50 mm long is resting on the HP on a side of its base. The axis of the solid makes an angle of 30° with the HP and 45° with VP. Draw the projections of the prism.
Q. 6	A right regular pentagonal pyramid of side 30 mm and height 50 mm is resting centrally on the top of a cube of 50 mm side. Draw the isometric projection of the combined solid.

DEPARTMENT OF MECHANICAL ENGINEERING

PUNJABI UNIVERSITY, PATIALA

END SEMESTER EXAMINATION, FEBRUARY-2022

B.Tech 1ST YEAR, GROUP B

MCE-151, ENGINEERING GRAPHICS

TIME ALLOWED: 2 HOURS **MAX. MARKS: 50**

NOTE: SECTION A IS COMPULSORY. ATTEMPT ANY THREE QUESTIONS FROM SECTION B.

	Section A
Q. 1	***All parts of this question are compulsory*** **Marks (2*10= 20)**
I.	A point is 30 mm below HP and in VP. The point lies in __________and __________ quadrants.
II.	A prism is solid having _________ end faces joined together by __________ surfaces. (Dissimilar, Similar: Rectangles, Triangles).
III.	What is the difference between the frustum of a solid and a truncated solid?
IV.	A line is parallel to VP and perpendicular to HP. Its point view will be obtained on the __________ plane.
V.	A straight line AB is inclined to HP and VP. The side view of the line shows __________. (True Length, Point View, Projected Length, None of these)
VI.	Differentiate between an apparent and a true section.
VII.	In third angle projection, the top view of an object lies _________ the front view. (Above, Below, on XY Line)
VIII.	A plane lamina is perpendicular to HP and inclined to VP. The _________ view of the lamina shows an edge view.
IX.	The development of a right circular cylinder is obtained by _____________ method.

X.	What is the difference between an isometric and non-isometric line.
	Section B (10 x 3 = 30) Attempt any three questions.
Q. 2	A line AB 90 mm long is inclined at 45° to HP and its top view makes an angle of 60° with XY line. The end A is in HP and 12 mm in front of VP. Draw its projections and find its true inclination with VP. Locate HT and VT.
Q. 3	A pentagonal pyramid of base side 40 mm and height 80 mm rests with its base in HP with an edge of the base making 30° to VP. It is cut by a section plane perpendicular to VP and inclined at 40° to HP bisecting the axis. Draw the front view, sectional top view and true shape of the section.
Q. 4	A right square pyramid of base side 40 mm and height 80 mm is resting on HP on its base with one of the base edges inclined at 30° to VP. It is cut by a cutting plane perpendicular to VP and inclined at 45° to HP, bisecting the axis of the solid. Draw elevation, sectional plan and develop the lateral surface of the solid.
Q. 5	A hexagonal prism of base side 30 mm and height 50 mm long is resting on the HP on a side of its base. The axis of the solid makes an angle of 30° with the HP and 45° with VP. Draw the projections of the prism.
Q. 6	A right regular pentagonal pyramid of side 30 mm and height 50 mm is resting centrally on the top of a cube of 50 mm side. Draw the isometric projection of the combined solid.